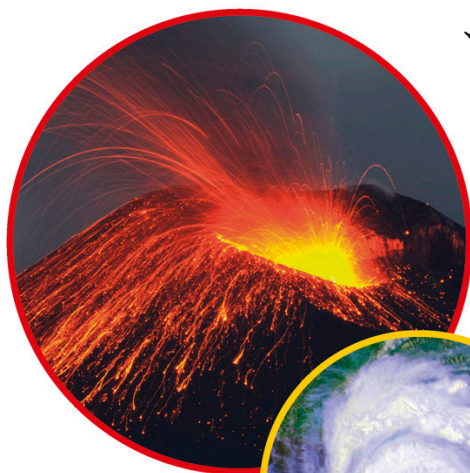




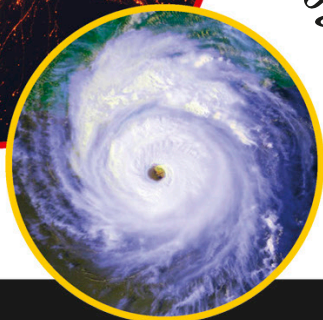
KEEPING the *balance*

There are two things that can have an extremely disruptive effect on an ecosystem—natural events and human activity. Natural events, such as a volcanic eruption, a flood, or a change in the climate, can change or destroy the habitat. The local species may be driven to look for a new home or killed because they can't escape. These events are usually catastrophic to the whole ecosystem, but leave it open to be colonized by a new set of species.

VOLCANIC ERUPTION



HURRICANE



Natural disasters...



HOLDING THE KEY

An ecosystem can often cope with losing one or two minor species, but there are some animals, called keystone species, that are essential to its survival. It is often a small predator that feeds on an herbivore which, in turn, eats the main plant in the environment. Without the predator the herbivore population grows out of control and crowds out other species.



Sea otters were once plentiful off the coast of California, but were hunted almost to extinction for their fur. Without the otters, the number of sea urchins, their favorite food, exploded. Sea urchins eat kelp (a giant seaweed), which shelters fish and feeds other herbivores. As the kelp disappeared, the whole ecosystem began to collapse.